Computational Mechanisms of Placebo Response in Major Depressive Disorder: A Reinforcement Learning Drift Diffusion Model Approach

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INTRODUCTION

- Anhedonia, a core symptom of major depressive disorder (MDD), represents a reduction in reward learning. The probabilistic reward task (PRT) quantifies this via response bias toward 'rich' (frequently rewarded) stimulus[1), 2), 3)].
- While MDD patients typically show blunted response bias, standard reinforcement learning (RL) models used to analyze this neglect the temporal dimension, relying solely on choice data [4)].
- The reinforcement learning drift diffusion model (RLDDM) integrates RL with the drift diffusion model (DDM) to simultaneously account for withing-trial decision processes and across-trial learning dynamics [5)].
- Extending previous work [3)], this study focuses on comparing the RLDDM against standard RL models to better capture the latent cognitive processes in PRT performance.

OBJECTIVES

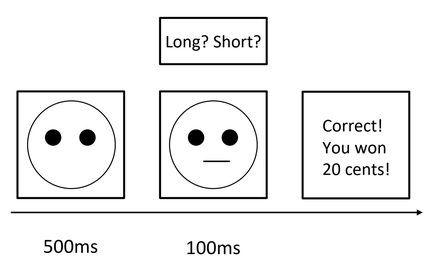
- Building a reinforcement learning (RL) model and reinforcement learning drift diffusion model (RLDDM) for probabilistic reward task (PRT)
- Comparing RL and RLDDM model by generating learning curve
- Investigating difference between placebo-responsive group and non-responsive group in major depressive disorder population using RLDDM framework

METHODS

- Participants & PRT

59 MDD patients

Placebo group Bupropion group



- Model Fitting

Parameters of all computational models were estimated based on hierarchical Bayesian framework. Posterior distributions were estimated using Markov chain Monte Carlo (MCMC) sampling. Models were built in Stan, and ran in R using rstan package.

- Reinforcement Learning Model

The RL model includes four parameters: instruction following bias (γ), stimulus uncertainty (ζ), reward sensitivity (ρ), and learning rate (ε) [1)].

- Reinforcement Learning Drift Diffusion Model

The RLDDM incorporates response time as an additional input and Drift Diffusion Model as a choice rule in RL model. RLDDM in this study includes five parameters: learning rate(ε), boundary separation (a), drift scaling parameter (v), non-decision time(τ), and starting point bias (β). [5), 6)]

- Model Comparison

For comparison between RL model and RLDDM, learning curves were generated from actual data and MCMC samples from the RL model and the RLDDM. Participants were expected to demonstrate faster learning for the rich stimulus than lean stimulus.

- Placebo Group Difference

Placebo-responsive group (N = 10)

Non-responsive group (N = 21)

- Signal Detection Theory

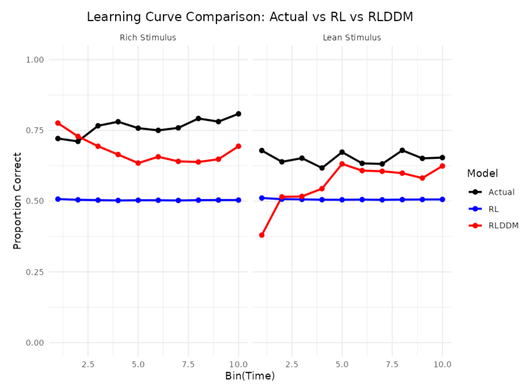
Sensitivity (d') and response bias (c) were calculated based on signal detection theory (SDT) to analyze placebo-responsive group's behavior [7)].

RESULTS

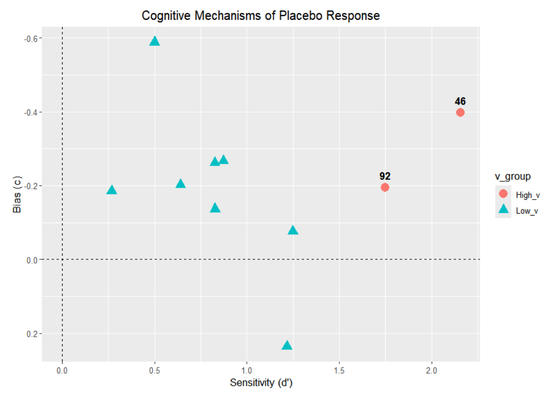
- RL Model Parameter Estimation

Parameters	RL Model (Mean [95% CI])	RLDDM (Mean [95% CI])
Learning rate (ε)	0.035 ([0.001, 0.126])	0.230 ([0.125, 0.360])
Instruction following bias (γ)	0.046 ([0.013, 0.095])	-
Stimulus uncertainty (ζ)	0.641 ([0.395, 0.885])	-
Reward sensitivity (ρ)	0.006 ([0.002, 0.013])	-
Boundary separation (a)	-	3.601 ([2.145, 7.986])
Drift rate scaling parameter (v)	-	1.163 ([0.150, 2.475])
Non-decision time (τ)	-	0.210 ([0.182, 0.303)]
Starting point bias (β)	-	0.869 ([0.781. 0.952])

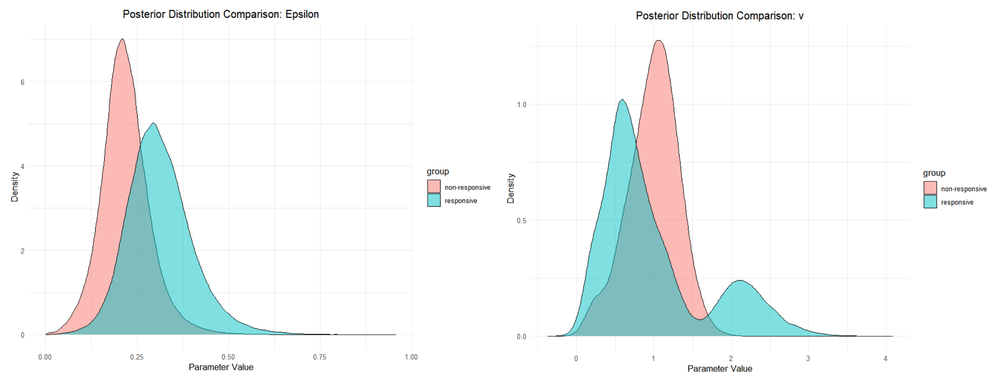
- Model Comparison: Learning Curves



- Within Placebo-responsive Group : Signal Detection Theory



- Placebo Group Difference: RLDDM



CONCLUSIONS

- Model Comparison:

RLDDM had more flexibility and tracked actual learning trajectory more closely.

- Placebo Group Difference: RLDDM

Placebo-responsive group exhibited higher posterior mean for learning rate, which suggests that placebo effect involves rapid update of reward belief. Posterior distribution of drift scaling parameter had bimodality, driven by two participants with faster evidence accumulation. This suggests possibility of placebo effect on cognitive domains.

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